Sympera AI: Home Assignment

Objective

Demonstrate your ability to bridge raw banking data with market context to drive growth opportunities for a Bank Relationship Manager (RM).

The Scenario

You are analyzing a small construction firm, 'Elite Builds LLC,' based in Utah. While the broader construction industry is facing 15% inflation in raw material costs, Elite Builds has remained consistently profitable.

Data Assets

1. Market Context (Snippet)

Q1 2026 Industry Report: Residential construction costs have surged by 15% due to supply chain volatility. Average net margins for mid-sized firms have dropped to 4%. Banks are tightening credit for firms showing high 'Expense Velocity'.

2. Banking Data (Last 20 Transactions)

Date	Description	Category	Amount	Type
2026-03-01	Client Payment - Project Alpha	Revenue	45000	Credit
2026-03-02	Vendor: Intermountain Lumber	Inventory	-12000	Debit
2026-03-04	Payroll Service	Operations	-8500	Debit
2026-03-05	Client Payment - Project Beta	Revenue	22000	Credit
2026-03-08	Lease Payment – Excavator	Equipment	-2200	Debit
2026-03-10	Gas & Fuel	Operations	-600	Debit
2026-03-12	Vendor: Steel Supply Co	Inventory	-4000	Debit

2026-03-15	Client Payment – Deposit	Revenue	15000	Credit
2026-03-18	Software Subscription (AutoCAD)	Tech/Growth	-150	Debit
2026-03-20	Marketing - Local SEO	Tech/Growth	-500	Debit
2026-03-22	Bonus Interest Earned	Interest	120	Credit
2026-03-23	Vendor: Intermountain Lumber	Inventory	-11000	Debit
2026-03-25	Payroll Service	Operations	-8500	Debit
2026-03-26	Internal Transfer to Savings	Savings	-5000	Debit
2026-03-27	Client Payment - Project Gamma	Revenue	30000	Credit
2026-03-28	Site Permit Fees	Operations	-450	Debit
2026-03-29	Vendor: Steel Supply Co	Inventory	-3800	Debit
2026-03-30	Tool Repair	Operations	-300	Debit
2026-03-31	Utility Bill	Operations	-400	Debit
2026-03-31	New Equipment Downpayment	Growth	-10000	Debit

The Challenge Tasks

Task 1: Positive Signal Engineering (Python)

Write a Python script to derive 'Positive Signals of Interest'. Identify at least two growth signals by joining the banking data with the market snippet. Look for 'Resilience Alpha'

(how this company is beating the 15% industry cost surge) and 'Growth Reinvestment' (identifying strategic outflows vs. operational noise).

Task 2: RAG Dataset Creation & Summarization

Develop a structured JSON object capturing segmentation insights for a RAG system. The output must be under 300 tokens. Include segments for behavioural (spending trends), demographics, and psychographics (brand affinity/lifestyle).

Task 3: Prompt Engineering & RM Action Plan

Submit the exact System Prompt used to generate the summary (Task 2). Demonstrate 'Persona-based' instruction and handle constraints like tone and format. Define a specific bank product recommendation and draft a 2-sentence 'Hook' for the Relationship Manager to use.

Submission Requirements

- Public GitHub Repository structure:
- - /scripts: Python extraction logic.
- - /output: Final customer_segments.json.
- - README.md: Overview of approach, assumptions, the 'RM Hook', and the exact System Prompt used.

Evaluation Criteria

- Business Insightfulness (40%): Quality of positive growth signals relative to market benchmarks.
- RAG Readiness (30%): Token efficiency and structured JSON quality.
- Technical Proficiency (20%): Clean, professional Python code.
- Clarity (10%): Quality of the README and documentation.